

Algebra First-Year Practice Exam A, Part 2

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let V be a finite-dimensional vector space over a field F and $T \in \text{End}_F(V)$ a linear transformation. Prove that there is a unique monic polynomial $m(x)$ in $F[x]$ with the property that $m(x)$ has the minimum degree among nonzero polynomials $f(x)$ such that $f(T)$ is the zero transformation.
2. Let $n \geq 1$ and let $K = \mathbb{Q}(\sqrt{1}, \sqrt{2}, \dots, \sqrt{n})$. Prove that $2^{1/3} \notin K$.
3. Let R be a commutative ring with $1 \neq 0$. Prove that R has a minimal prime ideal. (In other words, show that R contains a prime ideal P such that if $Q \subseteq P$ is a prime ideal of R , then $Q = P$.) You may assume that R contains a maximal ideal.
4. Prove that the ring $\mathbb{Z}[\sqrt{-2}] = \{a + b\sqrt{-2} : a, b \in \mathbb{Z}\}$ is a Euclidean domain with respect to the norm $N(a + b\sqrt{-2}) = a^2 + 2b^2$.
5. Say that an R -module M is torsion if for every $x \in M$ there is $r \in R$ such that $r \neq 0$ and $rx = 0$. The annihilator of M is defined to be

$$\text{Ann}(M) = \{s \in R : sx = 0 \text{ for all } x \in M\}.$$

Prove that if M is a finitely generated torsion module over an integral domain R , then $\text{Ann}(M) \neq \{0\}$. Give an example of a torsion R -module M over an integral domain R such that $\text{Ann}(M) = \{0\}$.